

Transformations of Trigonometric Functions Summary

-If $|a| > 1 \rightarrow$ vertical expansion by a factor of $|a|$

-If $0 < |a| < 1 \rightarrow$ vertical compression by a factor of $|a|$

-If $a < 0 \rightarrow$ reflection in the x -axis

Amplitude = $|a|$

-If $|k| > 1 \rightarrow$ horizontal compression by a factor of $\left|\frac{1}{k}\right|$

-If $0 < |k| < 1 \rightarrow$ horizontal expansion by a factor of $\left|\frac{1}{k}\right|$

-If $k < 0 \rightarrow$ reflection in the y -axis

New period = original period

$|k|$

$$y = a f[k(\theta - d)] + c$$

-If $d > 0 \rightarrow$ horizontal translation $|d|$ units to the right

-If $d < 0 \rightarrow$ horizontal translation $|d|$ units to the left

Phase Shift = $\pm d$ units

*(positive $\rightarrow$ translated to the right,
negative $\rightarrow$ translated to the left)*

-If $c > 0 \rightarrow$ vertical translation $|c|$ units up

-If $c < 0 \rightarrow$ vertical translation $|c|$ units down

Vertical Displacement = $\pm c$ units

*(positive $\rightarrow$ translated up,
negative $\rightarrow$ translated down)*

Note: Don't forget to factor the horizontal expansion/compression value before reading the horizontal translation. Ex: $y = \sin(2x + 6)$ should be factored into $y = \sin[2(x + 3)]$

Strategy in setting the scale:

In radians: Find the **LCD** between the phase shift and a quarter of the period.

For example: If the period is π , and the phase shift is $\frac{\pi}{3}$, find the LCD of $\frac{\pi}{4}$ and $\frac{\pi}{3}$. A scale of $\frac{\pi}{12}$ radians will be the easiest to work with.

In degrees: Find the **GCF** between the phase shift and a quarter of the period.

For example: If the period is 180° and the phase shift is 30° , find the LCD between 45° ($180^\circ \div 4$) and 30° . A scale of 15° will be the easiest to work with.